

Is Hyperbole a Scalar Inference?

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Abstract. This article is concerned with the interpretation of hyperbolic statements. In the first part, we contend that Kao, Wu, Bergen, and Goodman’s (2014)—to our knowledge, the first (and only) attempt to derive hyperbole interpretations using formal methods—is unsuccessful. In the second part, we put forth a conjecture—roughly, a sentence S can be interpreted hyperbolically as meaning S^- , where S^- is weaker than S , only if S^- is a scalar alternative of S . We substantiate this conjecture by a series of novel empirical observations.

1 Introduction

A hyperbole is an exaggerated statement that is not intended to be taken literally and is typically used to convey a positive or negative feeling, or express a positive or negative evaluation. For example, one may utter ‘Chomsky wrote 10,000,000 books’, despite being common ground that Chomsky didn’t write 10,000,000 books, to communicate that he wrote a lot of books and that one is positively impressed by this. To the best of our knowledge, Kao, Wu, Bergen, and Goodman (2014) stands alone as the only attempt to derive hyperbole interpretations using formal methods.¹ In this article, we argue that this attempt is unsuccessful. In addition, we advance a conjecture that, if proven true, would establish a fundamental connection between the interpretation of hyperbolic statements and the computation of scalar implicatures.

2 Kao, Wu, Bergen, and Goodman (2014)

Kao, Wu, Bergen, and Goodman (2014), which is couched within the Rational Speech Act (RSA) framework (Frank and Goodman 2012), has two distinctive features. First, it assumes that utterance interpretation operates along two dimensions (formalised as QUDs)—namely, the state-of-the-world dimension (e.g. *What’s the weather like?*, *How much money does Chris owe?*, *Is John coming the party tomorrow?*) and the speaker-affect dimension (*Is the speaker affected?*); in addition, it assumes that the speaker chooses her utterances to maximise the probability of accomplishing her goals, which can be either communicating information along

¹ Kao et al. (2014) is also presented as an account of ‘pragmatic halo’ effects in natural language (Lasersohn 1999). In this article, our sole focus is on Kao et al. (2014) as an account of hyperbole.

the state-of-the-world dimension, communicating information along the speaker-affect dimension, or communicating along both.

Bergen (2016: 145) provides an example—using as input sentence ‘Bob owes me \$1,000,000’—that illustrates the workings of this account. In Bergen’s (2016) example, there are 3 possible money states (\$10, \$100, and \$1,000,000) that correspond to the amount of money that Bob owes the speaker and 2 possible affect states ($\top$, which indicates negative affect, and $\perp$, which indicates absence of affect). There is a total of 6 worlds: each world consists of a money-affect pair ($M = x, A = y$), where x stands for the amount of money that Bob owes the speaker, while y stands for the speaker’s affect. The speaker is assumed to know exactly how much money Bob owes her as well as what her affective state is. The listener’s prior (which is also the common prior²) is taken to satisfy the following constraints: First, $P(A = \top \mid M = 10) < P(A = \top \mid M = 100) < P(A = \top \mid M = 1,000,000)$ (i.e. the greater the amount of money that Bob owes the speaker, the likelier it is that the speaker is unhappy). Second, $P(M = 1,000,000) < P(M = 100) \leq P(M = 10)$ (i.e. Bob is less likely to owe a million dollars than he is to owe smaller amounts of money). The values of the prior that Bergen uses in his simulation are given below.

x	$P(M = x)$	$P(A = \top \mid M = x)$
\$10	0.495	0.1
\$100	0.495	0.33
\$1000000	0.01	0.9

Table 1. The column labelled $P(M = x)$ provides the marginal probability of each possible money state, while the column labelled $P(A = \top \mid M = x)$ provides the conditional probability of the speaker having affect given each possible money state.

Bergen (2016) assumes that speaker is in world (100, $\top$), wants to communicate her affect (i.e. the speaker wants to address the QUD *Is the speaker affected?*) and, to do so, can utter either ‘(Bob owes me) \$10’, ‘(Bob owes me) \$100’, or ‘(Bob owes me) \$1,000,000’. According to Table 1, ‘\$1,000,000’ is the best that the speaker can do to accomplish this (because she knows that, upon hearing ‘\$1,000,000’, the listener will infer that the probability of her having affect is 0.9). Let’s now look at what happens on the listener’s side. The listener is uncertain as to whether the speaker wants to communicate her affect or, alternatively, state-of-the-world

² Like most game-theoretic models of communication, Kao et al. (2014) exploits the *common prior assumption*—for an overview, see Morris (1995).

information—for example, let’s assume that the listener is uncertain as to whether the speaker wants to address the QUD *Is the speaker affected?*, or alternatively, the QUD *How much money does Bob owe the speaker?* Under the former QUD, the utterance ‘Bob owes me \$1,000,000’ has the result of increasing the probability that the speaker is affected; under the latter QUD, this utterance indicates the precise amount that Bob owes the speaker. Since it is close to impossible that Bob owes the speaker \$1,000,000, the listener will conclude that the speaker must be intending to answer the speaker-affect QUD, and not the state-of-the-world QUD. That is, the listener will conclude that the speaker uttered S not because she thinks S is true but because she is affected (in this case, negatively).

This example illustrates the general idea that underlies Kao et al.’s (2014) account: To communicate her affect, a speaker can utter a declarative sentence S—even if she knows that S isn’t true—provided that (i) S is very likely to be false according to the common prior, and (ii) does a good job at communicating information along the speaker-affect dimension. In such conditions, upon hearing S, the listener will discard the hypothesis that the speaker wants to communicate that S is the case and will conclude instead that the speaker’s goal is to communicate her affect.

The listener, importantly, will typically also extract state-of-the-world information from hyperbolic statements; this, as Bergen (2016: 149) points out, ‘is because of the listener’s *a priori* knowledge that the state of the world and the speaker’s affect are likely to be linked.’ In the example discussed here, if the speaker has affect T (namely, negative affect), then it is a lot more likely that she is in a \$100-world than in a \$10-world—that is, it’s a lot more likely that she is in a world in which Bob owes her a significant amount of money than in a world in which Bob owes her little money.

2.1 A critical assessment of Kao et al.’s (2014) account

Kao et al.’s (2014) work represents a significant achievement in formal pragmatics: they managed to put together an explicit model of a phenomenon that had historically resisted formal analysis. However, upon examining the model’s predictions, at least three problems become apparent. In what follows, we discuss these problems in turn.

2.1.1 The water is boiling! (or the model simultaneously under- and over-predicts)

In plain English, Kao et al. (2014)—applied to ‘Bob owes me \$1,000,000’—could be summarised as follows:

(i) The speaker utters Bob owes me \$1,000,000. (ii) ‘This can’t be true!’, the listener says to herself. ‘The speaker is surely trying to communicate something else other than state-of-the-world information.’ (iii) ‘If the speaker was in a world in which Bob owes me \$1,000,000 was true’, the listener reasons, ‘she’d be, almost certainly, unhappy.’ (iv) ‘Voilà’, the listener concludes, ‘the speaker is trying to tell me she’s very unhappy (i.e. the speaker is trying to communicate her affect).’ (v) ‘And, surely, if the speaker is very unhappy, then Bob must owe her a significant amount.’

It’s worth noting that the reasoning in (i)-(v) works in highly idealised configurations in which the speaker’s affect state, if it can be explained at all, can be explained only by reference to a single state-of-the-world dimension ϕ —in Bergen’s example, the money-owed dimension; in Kao et al.’s (2014) simulations, the price dimension—and, furthermore, where there is only one likely reason for the speaker to be (negatively/positively) affected—namely, the world being somewhat high along ϕ —in Bergen’s example, the only likely reason for the speaker being negatively affected is Bob owing him a significant amount of money; in Kao et al.’s (2014) simulations, the only likely reason for the speaker being negatively affected is the price of certain item being too high.

In what follows, we would like to consider a slightly more complex case: a case in which, like in Kao et al. (2014), the speaker’s affect state is causally linked to just one state-of-the-world dimension ϕ but, unlike in Kao et al.’s (2014), there are *two* likely reasons that can account for the speaker’s affect state—namely, the world being somewhat high along ϕ *as well as* the world being somewhat low along ϕ . This, as will be shown, creates problem for the model.

Let’s consider the utterance ‘The water (in the bathtub) is boiling’: in most situations, it won’t convey the information that the water in the bathtub is literally boiling but rather will be interpreted hyperbolically—as meaning, roughly, ‘The water (in the bathtub) is very hot’. If one tries to run the reasoning in (i)-(v) with ‘The water (in the bathtub) is boiling’, however, a problem arises at step (v). Let’s see this:

(i) The speaker utters The water in the bathtub is boiling. (ii) ‘This can’t be true!’, the listener says to herself. ‘The speaker is surely trying to communicate something else other than state-of-the-world information.’ (iii) ‘If the speaker was in a world in which The water in the bathtub is boiling was true’, the listener

reasons, ‘she’d be, almost certainly, unhappy.’ (iv) ‘Voilà’, the listener concludes, ‘the speaker is trying to tell me she’s very unhappy (i.e. the speaker is trying to communicate her affect).’ (v) ‘And, surely, if the speaker is very unhappy, then *the bathtub water must be either too cold or too hot to take bath.*’

The problem is this: At step (v), the listener can only conclude the bathtub water must be either too cold or too hot to take bath (the unhappiness of the speaker may be due to the water being too hot but also to the water being too cold). But this is too weak a conclusion—in the real world, an utterance ‘The water (in the bathtub) is boiling’ gives rise to a much stronger inference, namely, *the water in the bathtub is very hot (too hot to take a bath).*

2.1.1.1 Testing Kao et al.’s (2014) model

To show that Kao et al.’s (2014) model does indeed fail in the configuration just described, let’s set the model’s various parameters and examine the predictions that it makes.

[I] Worlds

20 in total. Each world consists of a temperature-affect pair ($T = x$, $A = y$), where x stands for the temperature of the bathtub water, while y stands for the speaker’s affect state. The set of possible temperatures is $\{10^\circ\text{C} \times n : n = 1, 2, \dots, 10\}$, while the set of possible affect states is $\{0, 1\}$ —as in Kao et al. (2014), ‘0’ means without affect, ‘1’ means with affect, and affect, when present, is taken to be negative.

[II] Priors

The prior probabilities in Table 2 were used:

x	$P(T = x)$	$P(A = 1 \mid T = x)$
10°C	0.03	0.9
20°C	0.05	0.8
30°C	0.2	0.7
40°C	0.44	0.01
50°C	0.2	0.75
60°C	0.04	0.8
70°C	0.02	0.9
80°C	0.012	0.97
90°C	0.005	0.98
100°C	0.003	0.99

Table 2. The column labelled $P(T = x)$ provides the marginal probability of each possible temperature state, while the column labelled $P(A = 1 \mid T = x)$ provides the conditional probability of the speaker having affect given each possible temperature state.

The priors were chosen following three ‘reasonableness’ criteria:

- 40°C is (more or less) the ideal bath temperature (just above the human body temperature): at 30°C or below, the water is typically judged to be too cold for a bath, while at 50°C or above, too hot. Hence, we assumed that 40°C is, *a priori*, the most likely temperature (people are incentivised to seek comfort). As the temperature of the water moves away from the 40°C mark, either upwards or downwards, its probability decreases. (See Table 2, second column.)
- Since 40°C is the ideal temperature to have a bath, the probability of the speaker being affected is at its lowest at 40°C, and increases as the temperature moves away from this mark (see Table 2, third column.)
- The priors are not perfectly symmetric; this is intentional and is meant to reflect basic world-knowledge facts. For example, $P(T = 100^\circ\text{C}) < P(T = 10^\circ\text{C})$. This captures the fact that it’s more likely for the water to be at 10°C than at 100°C (bathtub water can’t reach boiling point in conventional households). Likewise, $P(A = 1 \mid T = 10^\circ\text{C}) > P(A = 1 \mid T = 100^\circ\text{C})$. This reflects the wide-spread preference to endure water at 10°C rather than at 100°C.

Finally, in order to avoid introducing bias towards either the cold or hot temperature spectrum, the values in Table 1 (second column) were chosen so that the likelihood of the water being too cold for a bath (that is, below 40°C) was equal to the likelihood of it being too hot (that is, above 40°C). As can be readily verified, $P(T = 10 \vee T = 20 \vee T = 30) = P(T = 50 \vee T = 60 \vee \dots \vee T = 100)$.

[III] Messages

5 in total, with the following meanings:³

- $\llbracket \text{too cold} \rrbracket = \{(T = x, A = y) : x < 40^\circ\text{C} \wedge y \leq 1\}$
- $\llbracket \text{OK} \rrbracket = \{(T = x, A = y) : x = 40^\circ\text{C} \wedge y \leq 1\}$
- $\llbracket \text{too hot} \rrbracket = \{(T = x, A = y) : x > 40^\circ\text{C} \wedge y \leq 1\}$
- $\llbracket \text{boiling} \rrbracket = \{(T = x, A = y) : x = 100^\circ\text{C} \wedge y \leq 1\}$
- $\llbracket \text{not OK} \rrbracket = \{(T = x, A = y) : x \neq 40^\circ\text{C} \wedge y \leq 1\}$

³ Each of these messages were assumed to have cost 0. This is in line with Kao et al. (2014: 12007): ‘[Our] analysis of hyperbole doesn’t involve utterance cost’.

[IV] QUDs

In Kao et al.’s (2014) model, as discussed, utterance interpretation operates along two dimensions (formalised as QUDs)—namely, the state-of-the-world dimension and the speaker-affect dimension. While the state-of-the-world dimension is represented by the QUD *What’s the price of x ?* (where x , depending on the simulation, is a watch, an electric kettle or a laptop), the speaker-affect dimension is represented by the QUD *Is the speaker affected?*

To represent the state-of-the-world dimension, we used a QUD structurally identical to the one used by Kao et al.’s (2014)—namely, *What’s the temperature of the bathtub water?* We shall refer to this QUD as the ‘fine-grained’ state-of-the-world QUD as each of its cells corresponds to a distinct temperature state. Unlike Kao et al. (2014), we also tested the model with ‘coarse-grained’ state-of-the-world QUDs—namely, *Is the bathtub water too hot, OK, or too cold?* and *Is the bathtub water too hot?* The former is a 3-cell partition (one cell containing those worlds in which the water temperature is above 40°C, one containing those worlds in which the water temperature is 40°C, and one containing those worlds in which the water temperature is below 40°C); the latter is a 2-cell partition (one cell containing those worlds in which the water temperature is above 40°C, and one containing those worlds in which the water temperature is 40°C or below). We did this to maximise the chances of the model getting things right—*a priori* it was conceivable that the model may fail when tested with the fine-grained QUD but succeed with one of the coarse-grained QUDs.

To represent the speaker-affect dimension, we used the same speaker-affect QUD that was used in Kao et al. (2014)—namely, a 2-cell partition such that one cell contains all the worlds in which the speaker has (negative) affect and one cell that contains all the worlds in which the speaker has no affect.

Kao et al. (2014), to allow for the possibility that the speaker may have the goal of communicating information along both the state-of-the-world and the affect dimension (as opposed to just having the goal of communicating state-of-the-world information, or just having the goal of communicating affect), included a third (‘hybrid’) QUD—namely, the conjunction of the state-of-the-world QUD and the speaker-affect QUD (see p. 12006, under *Materials and Methods*). We followed Kao et al. (2014) on this, but, in addition to running 3-QUD simulations (simulations that include the hybrid/conjunctive question), we also ran 2-QUD simulations (simulations containing just the state-of-the-world and the speaker-affect

QUDs). Once again, the rationale of this was maximising the chances of the model getting things right—*a priori*, it wasn’t implausible to think that the model might perform poorly when evaluated with three QUDs, yet do well when assessed with only two).

Table 3 shows the total number of simulations run (each row is a simulation, hence 6 simulations in total).

Simulation ID	Speaker-Affect QUD	State-of-the-World QUD	Affect & State QUD
1QQ	Is the speaker affected?	What’s the temperature of the bathtub water?	×
1QQQ	Is the speaker affected?	What’s the temperature of the bathtub water?	✓
2QQ	Is the speaker affected?	Is the bathtub water too cold, OK, or too hot?	×
2QQQ	Is the speaker affected?	Is the bathtub water too cold, OK, or too hot?	✓
3QQ	Is the speaker affected?	Is the bathtub water too hot?	×
3QQQ	Is the speaker affected?	Is the bathtub water too hot?	✓

Table 3. Full set of simulations ran. The QQ simulations use two QUDs, as indicated in the second and third column, whereas the QQQ simulations use three (they incorporate an additional ‘hybrid’ QUD that is the conjunction of the speaker-affect QUD and state-of-the-world QUD).

Simulations whose ID ends in ‘QQ’ only have two QUDs—the speaker-affect QUD, and one of the state-of-the-world QUDs; by contrast, simulations whose ID ends in ‘QQQ’ have three QUDs—namely, the affect QUD, one of the state-of-the-world QUDs, and the conjunction of the previous two QUDs. Thus, the only difference between n QQ and n QQQ is that the latter simulation has an extra QUD, i.e. the conjunction of the two QUDs associated with n QQ.

In each simulation, as in Kao et al. (2014), the prior probability over QUDs was assumed to be flat (in QQ-simulations each QUD had prior probability of $1/2$, whereas in QQQ-simulations, each QUD had a prior probability of $1/3$).

[V] Model’s predictions

Kao et al.’s (2014) theory of hyperbole, as mentioned in § 2, is couched within the RSA modelling framework. At the heart of this framework is the idea that speaker and listener recursively reason about each other’s intentions to arrive to pragmatically enriched

interpretations—for details about this and other aspects of RSA architecture, see Frank and Goodman (2012). Speaker and listener can, in principle, recurse to arbitrary depth.

For a given message or utterance u and recursive depth n , Kao et al.’s model (2014) generates an interpretation of u ; the interpretation of u at n is just the belief state of the n -level listener (formally, a probability distribution over worlds). Fig. 1 shows the interpretations that simulations 1QQ and 1QQQ assign to the utterance ‘The water (in the bathtub) is boiling’ at recursive depth 5:⁴

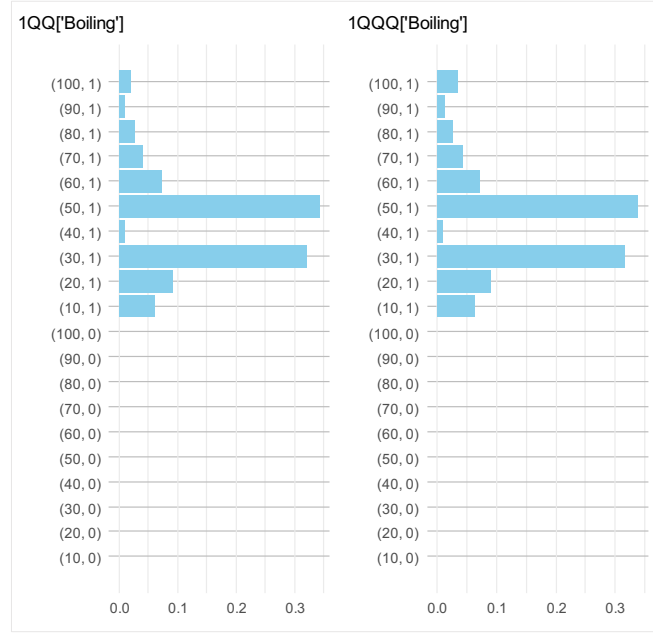


Fig 1. Interpretations for the utterance ‘The water (in the bathtub) is boiling’ according to 1QQ (left) and 1QQQ (right). Recursive depth 5, $\lambda = 20$.

As can be seen, simulations 1QQ and 1QQQ deliver nearly identical interpretations: after hearing ‘The water (in the bathtub) is boiling’, the listener comes to believe that the water must be either too hot (i.e. above 40°C) or too cold (i.e. below 40°C), and that the speaker is affected (note that the worlds that end in ‘1’ concentrate all the probability mass). Though the affect side of the inference is plausible, the state-of-the-world side isn’t: upon hearing ‘The water (in the bathtub)

⁴ Some technical remarks: At recursive depth 0, the model delivers literal interpretations, which we are not interested in. Non-literal interpretations (hyperbolic interpretation in this case) begin at recursive depth 1. We are illustrating the predictions of the model with level-5 interpretations, but we could have done so with level-1, level-4 or level-6 interpretation as well. To the extent that we were able to verify, all the interpretations obtained at recursive depths greater than 0 are similar enough to support our conclusions. It should also be noted that there is a parameter in the model (the ‘rationality’ parameter, often notated ‘ λ ’) that can affect the model’s output (as λ increases, the speaker becomes more rational, meaning the probability of choosing the optimal message given her goals gets closer to 1). In all our simulations, λ is set to 20, but other values could have been selected. We tested the model at lower (10) and higher (40) λ -values and did not find significant differences in the model’s predictions.

is boiling’, the listener should conclude that the water is very/extremely hot (too hot to take a bath), and not that the water may be too hot to take a bath but may also be too cold.

The same pattern emerges from the other simulations, as shown below.

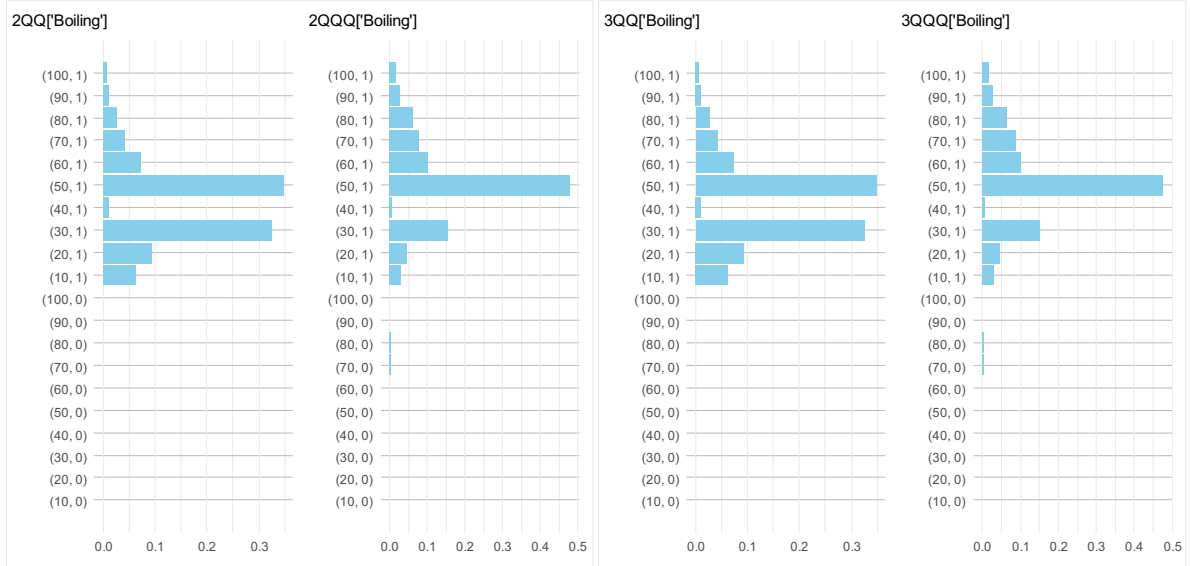


Fig 2. Interpretations for the utterance ‘The water (in the bathtub) is boiling’ according to 2QQ and 2QQQ (plot on left), and to 3QQ and 3QQ (plot on right). Recursive depth 5, $\lambda = 20$.

The only novelty is that in these simulations—which, unlike the ones in Fig. 1, use coarse-grained state-of-the-world QUDs (see Table 2)—there is a contrast between the QQ and the QQQ variants: indeed, quantitatively speaking, the QQQ variants (the 3-QUD simulations) are somewhat better than the QQ variants (the 2-QUD simulations). Take, for example, 2QQ and 2QQQ. In 2QQ the posterior probability of the water being too cold given an utterance of ‘The water (in the bathtub) is boiling’ is 0.481, while the probability of the water being too hot is 0.509. In 2QQQ, the probability tilts a bit towards the hot side of the spectrum (the ‘correct’ side of the spectrum): the posterior probability of the water being too cold given an utterance of ‘The water (in the bathtub) is boiling’ is 0.228, while the probability of the water being too hot is 0.763. Despite this relative improvement, the predictions that 2QQQ and 3QQQ make are still wrong: according to these simulations, after hearing ‘The water (in the bathtub) is boiling’, the listener comes to believe that the water, although it is more likely than not to be on the hot side of the spectrum, could also be too cold to take a bath.

Another way to assess Kao et al.’s (2014) predictions is to compare how the model interprets ‘The water (in the bathtub) is boiling’ vis-à-vis ‘The water (in the bathtub) is too hot’. Intuitively, one would like these two utterances to express the same (or very similar) state-of-

the-world (temperature) information: after all, in contexts in which ‘The water (in the bathtub) is boiling’ can be used hyperbolically, ‘The water (in the bathtub) is too hot’ could have been used instead. This isn’t the result that Kao et al. (2014) delivers, however. In Fig. 3 below, we plot the posterior distribution over temperature states (each temperature state is a set consisting of two worlds⁵) induced by ‘The water (in the bathtub) is boiling’ alongside the one induced by ‘The water (in the bathtub is) too hot’.

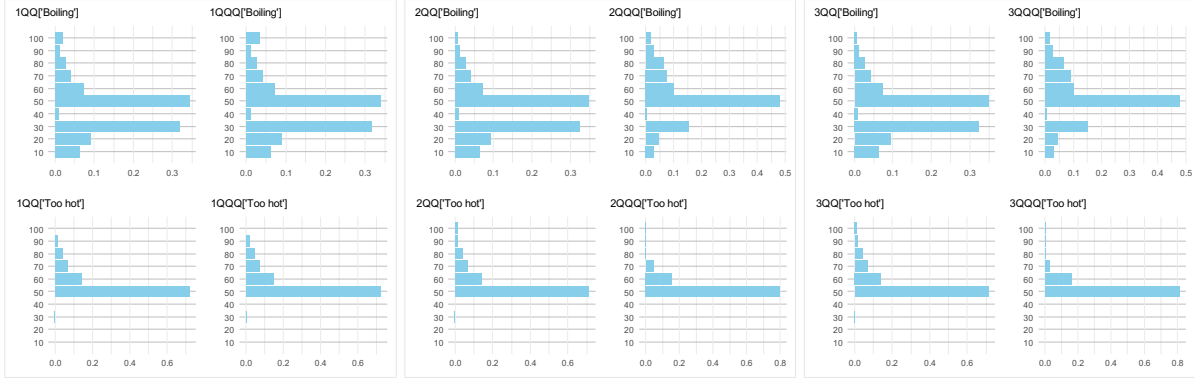


Fig 3. Posterior distributions over temperature states for the ‘The water (in the bathtub) is boiling’(first row) and for ‘The water (in the bathtub) is too hot’ (second row), as delivered by our six simulations (see Table 2). Recursive depth 5, $\lambda = 20$.

‘The water (in the bathtub is) too hot’ is interpreted as it should across all simulations—as shown in Fig. 3, virtually all, if not all, the probability mass lies in the too-hot-for-a-bath range (between 50°C and 100°C). The water (in the bathtub) is boiling’, by contrast, isn’t interpreted correctly: it induces a distribution according to which it is entirely possible for the water to be somewhere between 10°C and 40°C.

It should be noted that, in addition to failing to rule out the possibility of the water being too cold, the interpretation that the model assigns to ‘The water (in the bathtub) is boiling’ (irrespective of which simulations one picks) is problematic for at least two further reasons. As shown in Fig. 3, the listener (after interpreting ‘The water (in the bathtub) is boiling’) is predicted to come to believe that it is far more likely for the water to be in the 10-40°C range than at 60°C. This prediction doesn’t agree with intuitive judgment; if the water is (hyperbolically) boiling, then it’s too hot to take a bath; and that means that it’s not in the 10-40°C range but at 50°C or above. Similarly, the model predicts that, after interpreting ‘The water (in the bathtub) is boiling’, the listener will come to believe that the water is a lot more likely to be at 30°C than to be at 40°C. Again, this makes no intuitive sense; if the water is (hyperbolically) boiling, then it’s too hot to

⁵ For example, the 50° temperature state corresponds to the set $\{(T = 50, A = 0), (T = 50, A = 1)\}$.

take a bath; and, if that's the case, it's absurd to conclude that the water is more likely to be at 30°C (which is 20°C away from the beginning of the too-hot region) than at 40°C (which is only 10° away).

Given these results, we do not think that Kao et al.'s (2014) model can be seen as a successful account of hyperbole interpretation. The model is able to make intuitively correct predictions in highly idealised scenarios in which worlds have two dimensions and these dimensions are related in a particular way: as the state-of-the-world facts increase (e.g. the price of an item increases), the chances of the speaker being affected also increase (or at least do not decrease). However, in cases in which these dimensions exhibit a more complex relation (for example, in the boiling example just discussed), the model yields predictions that don't conform with intuitive judgment.

If our assessment of Kao et al.'s (2014) model is correct, it should be possible to 'fix' our simulations (that is, to get them to generate acceptable interpretations) simply by changing the relation between water-temperature states and the probability of the speaker being affected by such states. To verify this, we replaced the third column of Table 2 with the following values: 0.001, 0.001, 0.001, 0.001, 0.75, 0.80, 0.85, 0.90, 0.95, 0.99 (now, as the temperature goes up, the probability of the speaker having affect either goes up or stays the same). This adjustment did indeed fix the problem.⁶ In Fig. 4, we plot the posterior distribution over worlds for 1QQ and 1QQQ that result from running the model with the new 'friendly' priors vis-à-vis the posterior distribution over worlds for 1QQ and 1QQQ with the original priors (already plotted in Fig.1).⁷

⁶ Such priors are compatible, for example, with the following scenario: Imagine John wants to wash some items in the bathtub using special gloves. These gloves are made of a material that should not be exposed to temperatures of 50°C or higher. As the temperature rises above 50°C, the likelihood of the gloves melting and causing harm to John increases.

⁷ Virtually identical results are obtained for 2QQ/QQQ and 3QQ/QQQ.

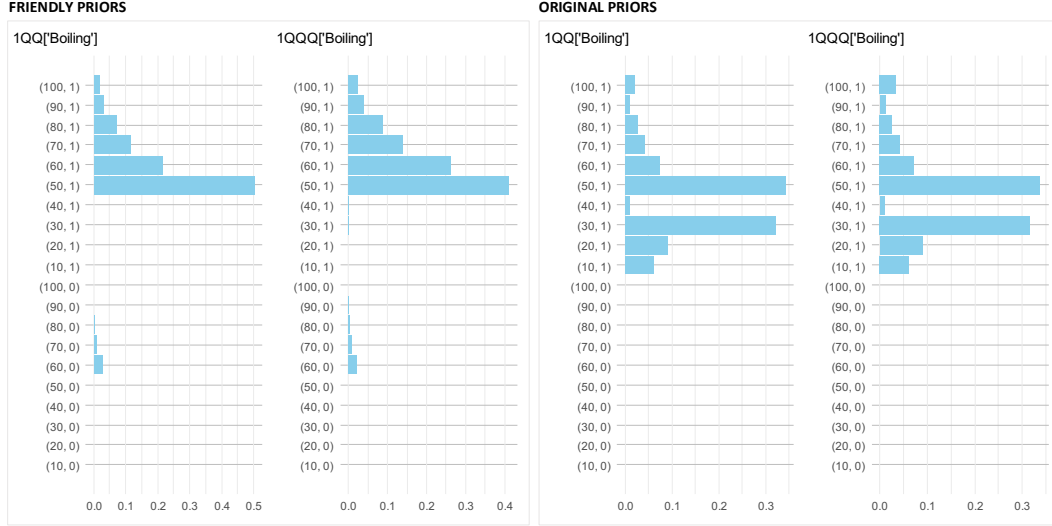


Fig. 4. Interpretations for ‘The water (in the bathtub) is boiling’ according to 1QQ and 1QQQ with ‘friendly’ priors (two plots on the left), and with the original priors (two plots on the right). Recursive depth 5, $\lambda = 20$.

With the new (friendly) priors, as shown in Fig. 4 (two plots on the left), the listener comes to believe, in both 1QQ and 1QQQ, that the water is too hot for a bath (the water is above 40°C), and that the speaker is (almost certainly) affected. These are *good* results, which are in sharp contrast with the results that are obtained with the original priors (see the two plots on the right, already given in Fig. 1).

To sum up, Kao et al. (2014) model, though it yields reasonable predictions in some configurations, doesn’t work in others. As we have seen, the model *underpredicts*—for example, it fails to generate the correct hyperbolic interpretation for ‘the water (in the bathtub) is boiling’ when fed with the (perfectly reasonable) priors in Table 2 (priors reflecting the fact that the speaker can be affected as a result of the water being too hot but also as the result being too cold). Furthermore, when fed with such priors, the model *overpredicts*—it predicts that ‘the water (in the bathtub) is boiling’ can be used to communicate that the water is either too cold or too hot to take bath, something which isn’t true.

2.1.2 King won the Nobel (or the model over-generates)

Kao et al.’s (2014) account is global in nature: as far as the uttered sentence goes, it sees nothing else but its literal meaning (i.e. a proposition). On this account, therefore, the following contrast can’t be explained:

- (1) a. This exercise is impossible to solve. (H ✓)
b. This exercise doesn't have a solution. (H X)

(1)a and (1)b are (arguably) truth-conditionally equivalent⁸; however, (1)a, but not (1)b, supports hyperbole.⁹

Furthermore, if one adopts Kao et al.'s (2014) perspective, it's not clear how to account for the fact that (2)a, but not (2)b, works as an exaggeration ((2)b is just a plain lie).

- (2) a. Stephen King won every (literary) prize. (H ✓)
b. Stephen King won the Nobel Prize (in literature). (H X)

Indeed, according to Kao et al. (2014), it should be possible to interpret (2)b hyperbolically—like (2)a, it is extremely unlikely to be true and evokes high-affect worlds.

To prove this point, one just needs to apply Kao et al.'s (2014) model to (2)a and (2)b. Let's therefore establish the model's input parameters and examine the predictions that it makes.

[I] Worlds

We will assume that there're 4 possible prizes: A, B, C (which are prestigious) and the Nobel Prize (noted 'N', the most prestigious of all). Each world consists of a pair ($Z = x, A = y$), where x stands for a possible prize combination—e.g. King won A and nothing else (noted 'Axxx'), King won A and B and nothing else (noted 'ABxx') — while y stands for a possible affect state (either '1', which indicates that the speaker has positive affect, or '0' which indicates that the speaker is not affected). Thus, there are a total of 32 worlds (i.e. there are 16 possible prize-combination states and two affect states).

⁸ Note that 'The exercise is impossible to solve, but it has a solution' sounds incoherent (unless, of course, one reads 'impossible to solve' as 'impossible to solve [for us/for a human being/for an AI/etc.]').

⁹ If the reported judgment isn't immediately clear, consider (i) and (ii). In (i), B's utterance 'it's impossible to solve' can clearly be understood hyperbolically (as meaning 'it's very difficult to solve'). In (ii), by contrast, B appears to contradict herself, which indicates that her utterance 'it doesn't have a solution' cannot be interpreted as meaning 'it's very difficult to solve'.

- (i) A: Prof. Smith is mad. The exercise he gave us is really difficult. I've been trying to solve it for 5 hours to no avail.
B: I know. Strop trying. It's impossible to solve. The good news are: my dad, who is a mathematician, knows how to do it. Shall I tell you?
- (ii) A: Prof. Smith is mad. The exercise he gave us is really difficult. I've been trying to solve it for 5 hours to no avail.
B: # I know. Strop trying. It doesn't have a solution. The good news are: my dad, who is a mathematician, knows how to do it. Shall I tell you?

[II] Priors

x	$P(Z = x)$	$P(A = 1 \mid Z = x)$
xxxx ¹⁰	0.0104	0.01
Axxx	0.06	0.2
xBxx	0.06	0.2
xxCx	0.06	0.2
xxxN	0.0002	0.99
ABxx	0.142	0.6
AxCx	0.142	0.6
AxxN	0.0002	0.995
xBCx	0.142	0.6
xBxN	0.0002	0.995
xxCN	0.0002	0.995
ABCx	0.382	0.875
ABxN	0.0002	0.9975
AxCN	0.0002	0.9975
xBCN	0.0002	0.9975
ABCN	0.0002	0.99875

Fig 4. The prior distribution. The column labelled $P(Z = x)$ provides the marginal probability of each possible temperature state, while the column labelled $P(A = 1 \mid Z = x)$ provides the conditional probability of the speaker having affect given each possible temperature state.

The priors were chosen based on following assumptions (which we take to be entirely reasonable):

- The probability of being in a world in which King won the Nobel Prize is extremely low—as far as anyone knows, he didn’t.
- The probability of being in a world in which King won no prize is low (it would be highly unusual if an author of such repute had never been awarded a literary prize).
- Given King’s fame and reputation, he is more likely to have won two or more prizes as opposed to just one.
- If someone were to learn that King won the Nobel Prize, she would be almost certainly impressed.
- If someone were to learn that King won all the prizes contained in a given set of prizes X plus an additional prize, she would be more likely to be impressed than if she were to learn that King won just the prizes contained in X .

¹⁰ As its name suggests, this is the state in which King won no prize.

[III] Messages

9 in total, with the following meanings:

- $\llbracket \text{every prize} \rrbracket = \{(Z = x, A = y) : x = \text{ABCN} \wedge y \leq 1\}$
- $\llbracket \text{the Nobel Prize} \rrbracket = \{(Z = x, A = y) : x \text{ has letter 'N'} \wedge y \leq 1\}$
- $\llbracket \text{prize A} \rrbracket = \{(Z = x, A = y) : x \text{ has letter 'A'} \wedge y \leq 1\}$
- $\llbracket \text{prize B} \rrbracket = \{(Z = x, A = y) : x \text{ has letter 'B'} \wedge y \leq 1\}$
- $\llbracket \text{prize C} \rrbracket = \{(Z = x, A = y) : x \text{ has letter 'C'} \wedge y \leq 1\}$
- $\llbracket \text{prize A \& B} \rrbracket = \{(Z = x, A = y) : x \text{ has letters 'A' and 'B'} \wedge y \leq 1\}$
- $\llbracket \text{prize B \& C} \rrbracket = \{(Z = x, A = y) : x \text{ has letters 'B' and 'C'} \wedge y \leq 1\}$
- $\llbracket \text{prize A \& C} \rrbracket = \{(Z = x, A = y) : x \text{ has letters 'A' and 'C'} \wedge y \leq 1\}$
- $\llbracket \text{prize A \& B \& C} \rrbracket = \{(Z = x, A = y) : x \text{ has letters 'A', 'B' and 'C'} \wedge y \leq 1\}$

[IV] QUDs

The speaker, as in the previous examples, wants to communicate information along the state-of-the world dimension and/or along the speaker-affect dimension. The state-of-the world dimension is identified with the QUD *What prizes has King been awarded?* (a 16-cell partition), and the speaker-affect dimension with the QUD *Is the speaker affected?* (a 2-cell partition). We ran two simulations—a QQ simulation (a simulation with the state-of-the-world and the speaker-affect QUDs) and a QQQ one (a simulation that also includes the conjunction of the aforementioned QUDs).

[IV] Model's predictions

As shown in Fig. 5, both 'King won every prize' and 'King won the Nobel Prize' were interpreted hyperbolically.¹¹

¹¹ The plotted interpretations correspond to recursive depth 5. The qualifications made in fn. 4 apply here as well.

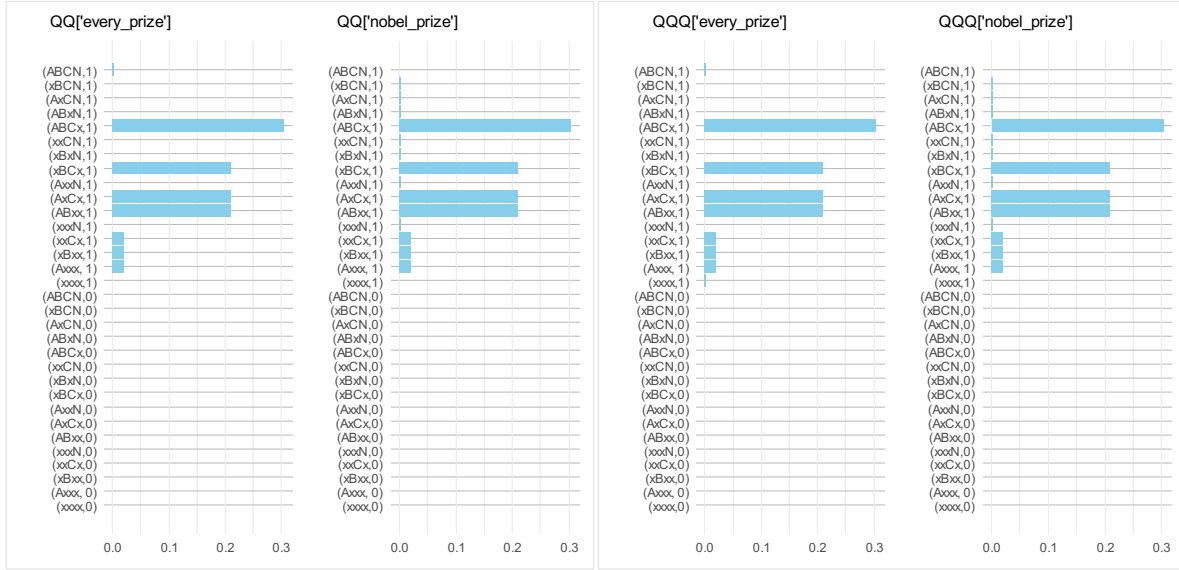


Fig 5. Final interpretations for the utterances ‘King won every prize’ and ‘King won the Nobel Prize’ according to QQ (two plots on left) and QQQ (two plots on right). Recursive depth 5, $\lambda = 20$.

Indeed, in both the QQ and the QQQ simulations, ‘King won every prize’ and ‘King won the Nobel Prize’ induced virtually the same posterior distribution, one which corresponds to a hyperbolic interpretation—i.e. the literal meaning of both of these sentences is false (all the worlds in which King won the Nobel Prize, including those in which King won every prize, have probability zero); the speaker is (positively) affected (the worlds that end in ‘1’ concentrate all the probability mass); King most certainly won a significant number of prizes (it is more likely than not that King won two or more prizes).

Thus, insofar as one accepts that ‘Stephen King won the Nobel Prize’ doesn’t work as an exaggeration (unlike ‘Stephen King won every prize’), then Kao et al.’s (2014) predictions for these utterances are obviously problematic. What the model appears to lack is a principle that distinguishes plain lies (e.g. ‘Stephen King won the Nobel Prize’, ‘Erdős proved the Riemann hypothesis’) from sentences that give rise to hyperbolic (non-literal) interpretation (e.g. ‘Stephen King won every prize’, ‘Erdős proved thousands of theorems’).

2.1.3 He solved an unsolvable problem! (or the model undegenerates)

Probabilistic conditioning is at the heart of Kao et al. (2014): the listener learns about the affective state of the speaker by conditioning the common prior on the proposition expressed by a factual utterance such as ‘The kettle costs \$10,000,000’. The following utterances, therefore, constitute a problem for this approach:

- (3) a. He broke an unbreakable chair. (*≈ He broke a chair that is very hard to break.*)
 b. He solved an unsolvable problem. (*≈ He solved a problem that is very hard to solve.*)

Indeed, (3)a-b, on any reasonable semantics, are necessary falsehoods (in probability talk, $P(\llbracket(3)a\rrbracket) = P(\llbracket(3)b\rrbracket) = 0$). Thus, if fed with (3)a-b, Kao et al.'s (2014) model will fail to derive the attested interpretations: it's just not possible to condition P on p in cases in which $P(p) = 0$ (in such cases, $P(\cdot|p)$ is undefined).

It might be argued that this problem isn't really Kao et al.'s (2014)—and, if it is, it is also everyone else's. Take, for example, the sentences '2 plus 2 equals 5' and '15678 divided by 3948 equals 4.2'. Given a possible world semantics, both sentences mean the same thing; however, 'The best math student from Harvard thinks that 2 plus 2 equals 5' strikes us as highly implausible, while 'The best math student from Harvard thinks that 15678 divided by 3948 equals 4.2' doesn't (she could have made a calculation mistake). Yet the underlined material has the same meaning (the underlined clauses are false in every possible world). Thus, one may argue that that Kao et al. (2014) is as limited as possible world semantics is: it's not fine-grained enough to successfully deal with necessary false sentences, or with sentences that have necessary falsehoods as sub-constituents.

That is one interpretation, and a defensible one.¹² But there is another interpretation: the real issue lies not with the inherent limitations of possible world semantics but rather with the model's architecture itself. Indeed, Kao et al.'s (2014) model is such that, to derive a non-literal interpretation (in this case, a hyperbolic interpretation), one must first update the common ground with the *literal* meaning of the utterance in question. This, however, is not the only possible architecture that a theory of hyperbole—and, more generally, a theory of non-literal interpretation—can have. (In what remains of this paper, in fact, we will explore a theoretical idea, which is instantiated in a possible world semantics, where the utterance's literal meaning plays a much less active role.)



Given the problems noted, we don't think that Kao et al.'s (2014) model can be regarded as a successful theory of hyperbole. At best, it's incomplete; at worst, it's incorrect. In the next

¹² Thanks to Leon Bergen for pressing on this point.

section, we will advance a theoretical conjecture, one that links the phenomenon of hyperbole with that of scalar implicatures, which could lead to the development of a new approach to hyperbole as well as to a refinement of Kao et al.'s (2014) account.

3 Is hyperbole a scalar inference?

Hyperbole can be viewed, at least in part, as a *weakening problem*, e.g. how come that from ‘Julia ate all of the cookies in the jar’ we can infer, under the right epistemic conditions, *Julia ate most of the cookies in the jar* and not, for example, *Julia ate all of the chocolate cookies in the jar*? (both interpretations are weaker than the original sentence, yet only the former is attested). Tellingly, scalar implicatures (SIs) give rise to a structurally analogous problem (only that, in this case, it’s a *strengthening problem*), e.g. how come that from ‘Julia ate some of the cookies in the jar’ we can infer *Julia ate some of the cookies in the jar, but it’s not the case that she ate all of the cookies in the jar* and not, for example, *Julia ate some of the cookies in the jar, but it’s not the case that she ate all of the chocolate cookies in the jar* (both interpretations are stronger than the original sentence, yet only the former is attested).

Most attempts at solving the problem that SIs give rise to¹³ have been guided by the following conjecture:¹⁴

- (4) A sentence S can be interpreted as meaning $S \wedge \neg S^+$, where S^+ is stronger (more informative) than S , only if S^+ is a scalar alternative of S .

From this perspective, the reason why ‘Julia ate some of the cookies in the jar’ cannot be interpreted as meaning ‘Julia ate some of the cookies in the jar, but it’s not the case that she ate all of the chocolate cookies in the jar’ but can be interpreted as meaning ‘Julia ate some of the cookies in the jar, but it’s not the case that she ate all of the cookies in the jar’ lies in the fact that ‘she ate all of the chocolate cookies in the jar’, unlike ‘she ate all of the cookies in the jar’, doesn’t qualify as a scalar alternative of the uttered sentence.

Here we would like to propose an analogous conjecture concerning the interpretation of hyperbolic statements:

¹³ A particular way of framing this problem came to be known as the ‘symmetry problem’ (Kroch 1972; Fox 2007; Katzir 2007, among others).

¹⁴ For a different approach to solving this problem, see, for example, Goodman and Stuhlmüller (2013).

- (5) A sentence S can be interpreted hyperbolically as meaning S^- , where S^- is weaker (less informative) than S , only if S^- (or a sentence contextually equivalent to S^-)¹⁵ is a scalar alternative of S .¹⁶

(5) enables us to answer the question raised at the outset of this section (how come that from ‘Julia ate all of the cookies in the jar’ we can infer, under the right epistemic conditions, *Julia ate most of the cookies in the jar* and not, for example, *Julia ate all of the chocolate cookies in the jar?*) in a straightforward manner: ‘Julia ate most of the cookies in the jar’ is a scalar alternative of ‘Julia ate all of the cookies in the jar’; ‘Julia ate all of the chocolate cookies in the jar’, by contrast, isn’t.

In addition, (5) makes two interesting predictions. Indeed, according to (5), if S can be interpreted hyperbolically as meaning S^- , then S must have a scalar implicature α that is weaker (less informative) than S and contextually equivalent to S^- . This leads us to expect the following: (i) it should be possible to paraphrase the hyperbolic interpretation that S gives rise to by using α (because α is contextually equivalent to S^-); (ii) if S is also a scalar alternative of α ,¹⁷ an utterance of α would be expected to have the scalar implicature $\neg S$ (because α is weaker than S). These predictions appear to be born out (see Fig. 6, next page).¹⁸

¹⁵ The parentheticals protect the conjecture from ‘silly’ counterexamples—for example, ‘No student is interested in the life of Dante’ can be interpreted hyperbolically as meaning ‘Few students are interested in the life of the author of the Divine Comedy’. The latter sentence, under standard assumptions, isn’t a scalar alternative of ‘No student is interested in the life of Dante’; however, ‘Few students are interested in the life of the author of the Divine Comedy’ is contextually equivalent to ‘Few students are interested in the life of Dante’, which is a scalar alternative of the uttered sentence.

¹⁶ We intend the expression *scalar alternative* to be understood in a theory-neutral way: an alternative that needs to be generated by some mechanism or other to derive a so-called scalar implicature.

¹⁷ In a framework such as Horn’s (1972), this is always the case, as the scalar-alternative relation is symmetric.

¹⁸ Of course, to arrive at a definite conclusion, a much larger dataset would need to be tested. Fig. 6 is only meant as a preliminary assessment of the proposed conjecture.

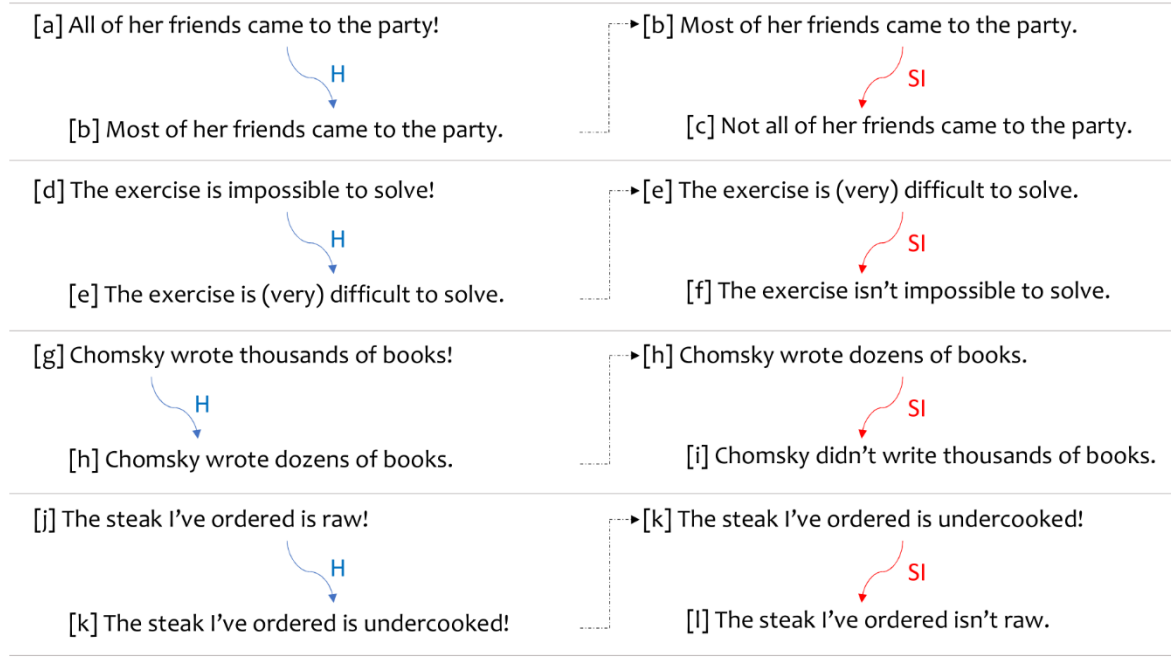


Fig. 6. The ‘H’ arrow associates a (false) statement S with α , a scalar alternative of S, whose meaning can be viewed as a possible hyperbolic interpretation of S. The ‘SI’ arrow associates α with $\neg S$.

More generally, if (5) holds, it’s reasonable to expect parallel behaviour between hyperbole and scalar implication, at least within certain paradigms. In what follows, we show that important parallelisms between these two phenomena can indeed be found.

3.1 Parallelism I (Numerals)

On neo-Gricean approaches to number interpretation, ‘Three boys *blah*’ is (typically) analysed as having weak truth-conditions (*Three or more boys blah*) and the upper-bounded component of the meaning (*No more than three boys blah*) as a scalar implicature (e.g. Horn 1972, Schulz and van Rooij 2006). From this perspective, the difference between (6)a and (6)b is that the former implicates that Chomsky didn’t write more than 10,000,000 books, whereas the latter entails it. (In what follows, we will assume that this analysis is correct.)

- (6) a. Chomsky wrote 10,000,000 books.
 $\neg_{SI}$ Chomsky didn’t write more than 10,000,000 books.
 b. Chomsky wrote exactly 10,000,000 books.
 $\neg_{SI}$ Chomsky didn’t write more than 10,000,000 books.
 $\Rightarrow$ Chomsky didn’t write more than 10,000,000 books.

Now, consider the sentences in (7) (*n* stands for a high yet plausible number in the context of book writing, e.g., 40 books): whereas (7)a supports hyperbole, (7)b doesn't.¹⁹

- (7) a. Chomsky wrote 10,000,000 books.
 $\neg_H$ Chomsky wrote *n* books or more.
 b. Chomsky wrote exactly 10,000,000 books.
 $\neg_H$ Chomsky wrote *n* books or more.

(6) and (7) can therefore be said to exhibit parallel patterns: 'Chomsky wrote 10,000,000 books', in its non-hyperbolic interpretation, implicates that Chomsky didn't write more than 10,000,000 books; 'Chomsky wrote exactly 10,000,000 books', by contrast, doesn't—indeed, this sentence entails, and hence doesn't implicate, that Chomsky didn't write more than 10,000,000 books. Similarly, 'Chomsky wrote 10,000,000 books' can be interpreted hyperbolically as meaning 'Chomsky wrote *n* books [or more]', where *n* is a high yet plausible number of books; 'Chomsky wrote exactly 10,000,000 books', by contrast, can't be interpreted in such a way. If one accepts the proposed conjecture and, furthermore, believes that the scalar alternatives of 'Chomsky wrote (exactly) 10,000,000 books' can be derived only by replacing '10,000,000' with another bare numeral, then the noted parallelism is unsurprising: 'Chomsky wrote exactly 10,000,000 books', unlike 'Chomsky wrote 10,000,000 books', has neither stronger nor weaker alternatives; hence 'Chomsky wrote exactly 10,000,000 books', unlike 'Chomsky wrote 10,000,000 books', is neither expected to be strengthened via a SI nor is it expected to be interpreted hyperbolically.

'Chomsky wrote *n* books [or more]', it should be noted, isn't the most natural paraphrase of (7)a's hyperbolic interpretation—as opposed to, say, 'Chomsky wrote a lot books', which feels more appropriate. Under the proposed conjecture, this is unexpected: 'Chomsky wrote *n* books [or more]' is a scalar alternative of (7)a; 'Chomsky wrote a lot books', by contrast, is not a

¹⁹ If the judgment is not immediately clear, consider the contrast between (iii) and (iv):

- (iii) A: How many books did Chomsky write?
 B: I don't know. The man wrote 10,000,000 books.
 (iv) A: How many books did Chomsky write?
 B: #I don't know. The man wrote exactly 10,000,000 books.

(iii)-B is clearly interpreted hyperbolically, as meaning 'I don't know. The man wrote a huge number of books.' (iv)-B, by contrast, doesn't lend itself to this interpretation and is understood literally instead—indeed, unlike (iii)-B, (iv)-B is perceived as incoherent.

scalar alternative of (7)a (at least not under standard assumptions), nor is it contextually equivalent to a scalar alternative of (7)a.²⁰

Does this observation threaten the proposed conjecture? We don't think it does, at least not in a fundamental way. If one turns the proposed conjecture into a procedural account of how the listener recovers the message that the speaker intends to communicate by means of a hyperbolic statement, one obtains something like the following: interpreting a sentence *S* hyperbolically amounts to identifying *S*'s semantic content not with $\llbracket S \rrbracket$ but with $\llbracket S^- \rrbracket$, where S^- stands for a scalar alternative of *S* that is weaker than *S* (and plausibly true given common knowledge). Upon hearing 'Chomsky wrote 10,000,000 books', however, interpreters are bound to have a *number* of possible landing sites for the hyperbole inference, each of which corresponding to a scalar alternative of the uttered sentence (e.g. 'Chomsky wrote 40 books [or more]', 'Chomsky wrote 41 books [or more]', 'Chomsky wrote 42 books [or more]', etc.). But which one to pick? In a realistic model of communication, the hyperbolic interpretation of a sentence like (7)a would need to be identified not with a scalar-alternative meaning (a proposition) but with a *distribution* over possible scalar-alternative meanings. From this perspective, it does make sense that, to describe the observed interpretation of (7)a, one feels compelled to use a *vague* sentence (e.g. 'Chomsky wrote a lot books', 'Chomsky wrote a vast amount of books'): these sentences manage to convey what an individual scalar alternative wouldn't be able to: the fact that, after interpreting (7)a hyperbolically, one can't tell with certainty what the lowest number of books that Chomsky definitely wrote is.²¹

3.2 Parallelism II (Hurford Disjunctions)

According to Hurford's (1974) well-known generalisation, often referred to as Hurford's constraint (HC), a sentence that contains a disjunctive phrase is odd if one of the disjuncts entails the other. This generalisation correctly predicts (8)a-b to be odd; however, it fails to make sense of the fact that (8)c is oddness-free ('all of the books' entails 'some of the books').

- (8) a. # John bought a computer or a laptop.
b. # John either killed or murdered Paul.
c. Jane read either some of the books or all of the books.

²⁰ Indeed, unless one picks a highly contrived context, 'a lot of books', unlike '*n* or more', would be expected to have a vague lower-bound.

²¹ For the connection between vague sentences and the communication of probabilistic information, see Égré et al. (2023).

Chierchia, Fox, and Spector (2008) argued that (8)c isn't a counterexample to HC: according to them, (8)c and other apparent violations of this generalisation involve the presence of an implicature-computing operator *exh* within the first disjunct that 'breaks' the entailment relation between the disjuncts—due to *exh*, (8)b ends up having the reading 'Jane read either some *but not all* of the books or all of the books'.

Now, one may ask: why isn't (8)a and (8)b also rescued by *exh*? This question, to our knowledge, has not yet received a conclusive answer. What one may call the 'received' view has it that the *is-an-alternative-of* relation is mediated by the notion of *lexical scale* (Horn 1972, Gazdar 1979) (hence the name '*scalar alternatives*'): whereas certain words (e.g. 'some'/'all') form a *lexical scale* (and hence are alternatives of one another), others (e.g. 'computer'/'laptop', 'kill'/'murder') do not. For this to constitute an answer to the question raised, however, we would need a general and reliable method capable to determine which pairs of words count, and which do not count, as a lexical scales, something we currently lack (recall Gazdar (1979:58) famous statement: "scales are, in some sense, 'given to us'").²²

Here we won't be concerned with this question; we'll simply take the contrast between (8)a-b and (8)c to be a manifestation of the following generalisation: if β a-entails α and ' α or β ' is felicitous, then β can be viewed as a scalar alternative of α ; conversely, if β a-entails α and ' α or β ' is infelicitous, then β cannot be viewed as a scalar alternative of α . Then, if (5), the proposed conjecture, is correct, the following prediction can be made: for any 'bad' Hurford

²² Katzir (2007), notably, takes a different approach and puts forth an explicit account of alternative generation based on considerations of complexity. Though this work makes sense of a great deal of data that was left unexplained in Horn's (1972) framework, it still fails to offer a principled explanation of the contrast between (8)c and (8)b, for example. For Katzir (2007), (8)c is good because 'all' is an alternative of 'some' (they are of the same complexity) while 'some but not all' (the so-called *symmetric* alternative) isn't (it's not of the same complexity as 'some'). Due to this, local exhaustification can proceed and Hurford constraint ends up not being violated. But this analysis leads to the prediction that (8)b should be good too: 'murder' is an alternative of 'kill' (they are of the same complexity) while 'unintentionally kill' (the symmetric alternative) isn't (it's more complex than 'kill'). Even if one were to endorse the view that 'murder' is represented at LF as 'intentionally kill', Katzir's account wouldn't be out of trouble: for one thing, it's not clear that such an assumption helps—according to Katzir (2007), a complex expression can serve as an alternative to a simpler expression provided the former is a linguistic constituent (and 'murder' is a linguistic constituent in (8)b; see Katzir's (2007) solution to Matsumoto's problem); furthermore, if 'murder' were to be represented at LF as 'intentionally kill', then 'John didn't murder Paul' would be predicted to implicate that John killed Paul—this implicature, however, doesn't seem to be there, as the contrast between (v) and (vi) reveals:

- (v) A: John didn't intentionally kill Paul.
B: Wait... Has John killed Paul?! I thought Jane had killed him.
- (vi) A: John didn't murder Paul.
B: # Wait... Has John killed Paul?! I thought Jane had killed him.

(v)-A clearly gives rise to the inference that John killed Paul, hence the naturalness of (v)-B; (vi)-A, by contrast, doesn't seem to give rise to the same inference—if it did, (iv)-B should be as felicitous as (v)-B, and it clearly isn't.

disjunction H , it is never possible to use H 's strong disjunct to hyperbolically convey that H 's weak disjunct is the case. This prediction, as illustrated in Fig. 7, is borne out.

Bad Hurford disjunctions	Hyperbole unavailable
[a] # Has John eaten a piece of fruit or an apple ?	[b] John has eaten an apple . [c] John has eaten a piece of fruit .
[d] # John bought a computer or a laptop .	[e] John bought a laptop . [f] John bought a computer .
[g] # John either killed or murdered Paul.	[h] John murdered Paul. [i] John killed Paul.

Fig. 7. Testing the following prediction: for any ‘bad’ Hurford disjunction H , it is never possible to use H 's strong disjunct to hyperbolically convey that H 's weak disjunct is the case.

It should be noted that, if one takes ‘good’ Hurford disjunctions as reference point, the data aren’t entirely parallel: though it’s possible to find good Hurford disjunctions whose strong disjunct can be used to hyperbolically to convey the weak disjunct, it’s also possible to find good Hurford disjunctions that don’t permit this, as illustrated below:

Good Hurford disjunctions	Hyperbole available
[a] ✓ Is this exercise (very) difficult or impossible ?	[b] This exercise is impossible . [c] This exercise is (very) difficult .
[d] ✓ Don’t eat beef that is (very) undercook or raw .	[e] The beef is raw . [f] The beef is (very) undercooked .
[g] ✓ John read most or all the books.	[h] John read all the books. [i] John read most of the books.
	Hyperbole unavailable
[j] ✓ John read 1 or 2 books.	[k] John read 2 books. [l] John read 1 book.
[m] ✓ John read the first two chapters or all the chapters .	[n] John read all the chapters . [o] John read the first two chapters .

Fig. 8. ‘Good’ Hurford disjunctions whose stronger disjunct gives rise to hyperbole ([a], [d], [g]) and ‘good’ Hurford disjunctions whose stronger disjunct does not give rise to hyperbole ([j], [m]).

The fact that this is so, it should be noted, doesn’t threaten (5), according to which a sentence S can be interpreted hyperbolically as meaning S^- *only if* (and not *if and only if*) S^- —or a sentence contextually equivalent to S^- —is a scalar alternative of S . The kind of example that would threat (5) is a ‘bad’ Hurford disjunction such that its strong disjunct can be used

hyperbolically to convey that its weak disjunct is the case—such cases, as far as we can tell, do not exist (see Fig. 7).

3.3 Parallelism III (Indirect Implicatures)

As discussed in §2.1.2., (1)a—repeated as (9)a below—can be used/interpreted hyperbolically; (1)b—repeated as (9)b below—, by contrast, cannot.

- (9) a. This exercise is impossible to solve.
 $\neg_H$ This exercise is (very) difficult to solve.
 b. This exercise doesn't have a solution.
 $\neg_H$ This exercise is (very) difficult to solve.

Now, consider (10)a-b—the negations of (9)a-b, respectively:

- (10)a. This exercise isn't impossible to solve.
 $\neg_{SI}$ This exercise is (very) difficult to solve.
 b. This exercise has a solution.
 $\neg_{SI}$ This exercise is (very) difficult to solve.

(10)a carries the (indirect) implicature that the exercise is difficult (or very difficult) to solve; (10)b, by contrast, doesn't. The observed parallelism, if viewed through the lens of the proposed conjecture, is not at all surprising: (10)a, and not (10)b, is interpreted as meaning 'The exercise isn't impossible but it's (very) difficult to solve' because (10)a, unlike (10)b, has the alternative 'The exercise isn't (very) difficult to solve'; likewise, (9)a, and not (9)b, is interpreted hyperbolically as meaning 'The exercise is (very) difficult to solve' because (9)a, unlike (9)b, has the alternative 'The exercise is (very) difficult to solve'.

At this point, a careful reader may feel compelled to make the following point:

Careful reader's point: Arguably, the most accurate paraphrase of the hyperbolic interpretation of (9)a is 'The exercise is *very* difficult to solve' (or 'The exercise is *extremely* difficult to solve') and not the plain 'The exercise is difficult to solve'. The author writes as if 'The exercise is *very* difficult to solve' was (or could be) a scalar alternative of 'The exercise is impossible to solve'; this assumption, however, is non-kosher: under standard assumptions, alternatives cannot be generated by replacing a lexical item (e.g. 'impossible') by a non-lexical constituent (e.g. 'very difficult')—cf. Horn (1972) and Katzir (2007). One can of course reject this

principle; but, if one does, then something needs to be said about the symmetry problem: indeed, if alternatives can be generated by replacing lexical items with non-lexical constituents, then how come that a sentence of the form ‘some *blah*’ appears to have ‘all *blah*’, but not ‘some but not all *blah*’, as an alternative?

One can approach this challenge in at least two ways. One option is to bite the bullet: if the proposed conjecture is correct, (9)a can be interpreted hyperbolically as meaning ‘The exercise is difficult to solve’ (whatever this means in the context in which (9)a is uttered²³) but not as ‘The exercise is very difficult to solve’. How to account, then, for the fact that an utterance of (9)a, at least intuitively, conveys not just that the exercise is difficult but that it is very (or extremely) difficult? This, it seems to me, is a possible answer:

- (i) An utterance of ‘The exercise is impossible’ leads the listener to conclude that the speaker has, or almost certainly has, negative affect. (How to derive this inference is an open question; Kao et al.’s (2014) model do derive this inference but, as discussed, they pay a high price for it, which raises the question of whether their method is viable in the general case.)
- (ii) If the speaker has negative affect, and this situation can be attributed to the exercise being difficult to solve, then she most likely wasn’t able to solve the exercise (or she did solve it but struggled in the process).
- (iii) Thus, it’s more likely than not that the speaker deems the exercise to be very difficult as opposed to difficult but not too difficult.

On this view, the inference that the exercise must be really/very/extremely difficult is a probabilistic inference: it’s more likely than not that that’s the case given something we’ve learned about the speaker (she is negatively affected by the exercise being difficult to solve).

Another option would be to reject the idea that ‘The exercise is very difficult to solve’ can’t be a scalar alternative of (9)a. Now, if one does this, something needs to be said—as the careful reader points out—about the so-called ‘symmetry problem’. And something can be said, we believe.

²³ As is known, relative adjectives like ‘difficult’, ‘tall’ or ‘rich’ express properties only relative to contextually-provided thresholds or comparison classes.

In recent years, the case has been made that non-monotonic expressions are, in a sense, more complex than monotonic ones: on the experimental front, it has been shown that non-monotonic quantifiers are harder to learn than monotonic ones (e.g. Steinert-Threlkeld and Szymanik 2017; Chemla et al. 2019); on the theoretical front, ways of measuring semantic complexity have been identified under which nonmonotonic quantifiers come out more complex than monotonic ones (e.g. Carcassi, Schouwstra, and Kirby 2019; van de Pol et al. 2023).

In the light of these findings, it seems reasonable to consider the following solution to the symmetry problem: ‘some *blah*’ has ‘all *blah*’ as an alternative because ‘some’ and ‘all’ are of the same *semantic* complexity (in the sense that they are both monotonic); likewise, ‘some *blah*’ doesn’t have ‘some but not all *blah*’ as an alternative because ‘some’ and ‘some but not all’ are not of the same complexity (the former is monotonic while the latter is non-monotonic).²⁴ Notably, if one adopts this solution, ‘The exercise is very difficult to solve’ can be (safely) characterised as a scalar alternative of (9)a: indeed, the expressions ‘difficult’ and ‘very difficult’, despite differing in word count, are both monotonic,²⁵ and can thus be considered equal in (semantic) complexity.²⁶

3.4 Comparison with Kao et al. (2014)

In § 2.1, we criticised Kao et al.’s (2014) framework on the basis of three data points, namely:

²⁴ This, we believe, is an attractive idea, an idea which Buccola, Križ, and Chemla (2022) appear to hint at. At the very least, it seems like a direction worth exploring, given that the general method of ‘breaking symmetry’ via expression complexity (i.e. ‘some but not all’, unlike ‘all’, has more words than ‘some’) faces non-trivial empirical challenges—see, for example, Breheny et al. (2018) and Buccola, Križ, and Chemla (2022).

²⁵ If an object *x* falls in the extension of ‘(very) difficult’, then any object *y* such that *y* is equally or more difficult than *x*, will also fall in the extension of ‘(very) difficult’. Because of this, ‘(very) difficult’ can be said to denote a monotonically increasing function.

²⁶ Note that the account just outlined immediately makes sense of the contrast between (vii) and (viii):

- (vii) Is this exercise difficult, or is it very difficult?
(It reads as ‘Is this exercise difficult but not very difficult, or is it very difficult?’ This is consistent with the view that ‘very difficult’ is a scalar alternative of ‘difficult’.)
- (viii) # Is this exercise difficult, or is it difficult but not very difficult?
(It doesn’t read as ‘Is this exercise very difficult, or is it difficult but not very difficult?’ This is the reading that one would expect if ‘difficult but not very difficult’ was a scalar alternative of ‘difficult’.)

(11) a. The water in the bathtub is boiling.

$\neg_H$ b. The water in the bathtub is (very) hot (i.e. too hot for a bath).

$\neg_H$ c. The water in the bathtub is either too hot or too cold (i.e. not at an appropriate temperature for taking a bath).

(12) a. Stephen King won every (literary) prize.

$\neg_H$ b. Stephen King won many (literary) prizes.

c. Stephen King won the Nobel Prize (in literature).

$\neg_H$ d. Stephen King won many (literary) prizes.

(13) a. He solved an unsolvable problem.

b. $\neg_H$ He solved a (very) hard/difficult problem.

Let's start with (11). Given an utterance of (11), Kao et al.'s (2014) model predicts an interpretation that (at least in part²⁷) can be paraphrased with (11)c while failing to derive the intuitively correct interpretation, paraphrased with (11)b. If the conjecture here proposed is on the right track, (11)a, under no circumstances, could be hyperbolically interpreted as (11)c: no approach to alternatives that we are aware of would predict 'too cold or too hot' to be in competition with 'hot'—it's not only more complex than 'hot' but it's also non-monotonic. By contrast, on the proposed framework, there would be no obstacle in assigning (11)a the hyperbolic interpretation in (11)b: 'hot' and 'boiling' are scalar competitors (as shown, for example, by the felicity of 'is the water hot or is it boiling?'); thus, (11)b can be viewed as an alternative of (11)a.

Let's now move to (12). As discussed, Kao et al.'s (2014) model predicts that both (12)a and (12)c can be understood as conveying that King won many literary prizes (this is a bad prediction). This prediction doesn't follow from the conjecture proposed here: (12)d is not a scalar alternative of (12)b (as shown, for example, by the *infelicity* of 'Either King won many literary prizes or he won the Nobel prize'); thus, on the proposed conjecture, (12)c cannot be hyperbolically interpreted as (12)d. Conversely, the fact that (12)a can be understood as (12)b is compatible with our proposal: (12)b is scalar alternative of (12)a (as shown, for example, by the felicity of 'Either King won many literary prizes or he won every literary prize').

²⁷ We say 'at least in part' because the model predicts that the listener will also extract from (11) information about the speaker's affect—namely, the information that the speaker is negatively affected.

Finally, let's take a look at (13)a. This example, as discussed, constitutes a problem for Kao et al.'s (2014) account: though (13)a can be used hyperbolically (to convey, roughly, that (13)b is true), Kao et al. (2014) is unable to generate this interpretation. As discussed in § 2.1.3, the issue is that, to derive a hyperbolic interpretation using Kao et al.'s (2014) method, one first needs to update the common prior with the literal meaning of the uttered sentence. This operation, however, can't be executed in the case of (13)a, because $P(\llbracket(13)a\rrbracket) = 0$ ((13)a is a necessary falsehood).

From the perspective of the proposed conjecture, the fact that (13)a is a necessary falsehood is irrelevant: (13)a can be understood as (13)b (irrespective of whether (13)a is or isn't a necessary falsehood) because (13)b is a scalar alternative of (13)a ('hard' and 'unsolvable' are scalar competitors, as illustrated, for example, by the felicity of 'Is the problem hard or is it unsolvable?'). To put this in more concrete terms, consider a simple procedural account of (13)a compatible with our conjecture according to which interpreting (13)a hyperbolically simply amounts to substituting 'unsolvable' by 'hard' (or by 'very hard', assuming that the addition of a modifier such as 'very' preserves the scalar-competitor relation) at LF. Such an account would successfully derive (13)b, and it would do so without updating the common ground with (13)a.

Now, the fact that the data in (11), (12), and (13) doesn't challenge the proposed conjecture doesn't mean that the proposed conjecture explains these data. The proposed conjecture is nothing else but a generalisation, and generalisations are not theories. By contrast, Kao et al.'s (2014) is a theory, one that strives to provide an explanatory account of hyperbole. Thus, we wouldn't want the proposed conjecture to be viewed as an alternative to Kao et al.'s (2014) account. It is not. RSA theorists, in fact, could attempt to refine Kao et al.'s (2014) model to capture our conjecture/generalisation (and avoid at least some of the problematic predictions discussed in § 2.1). This would involve, presumably, revising Kao et al.'s (2014) architecture to ensure that hyperbole interpretation is constrained by the (un)availability of scalar alternatives. How to accomplish this technically, assuming it is feasible, is something that we leave open for future work.

3.5 Caveats and limitations

The first thing that we would like to note is this: the proposed conjecture does contribute to solving what we have termed ‘hyperbole’s weakening problem’, but it does so only *partially*. Take, for example,

(14) Julia ate all of the cookies in the jar, and she watched her favourite movie.

(14) can be hyperbolically interpreted as meaning (i) ‘Julia ate *most* of the cookies in the jar (, and she watched her favourite movie)’ but not as meaning (ii) ‘Julia ate all of the cookies in the jar *or* she had a cup of coffee’. Note that both (i) and (ii) are scalar alternatives of (14), and are weaker than (14); the question raised in §3 reemerges: why is it that (14) can be interpreted hyperbolically as meaning (i) but not as meaning (ii)? On this question, the proposed conjecture is mute.

It seems clear that not all alternatives can serve as landing site for the hyperbole inference (e.g. ‘*p* and *q*’ cannot be hyperbolically interpreted as meaning ‘*p* or *q*’); in other words, being an alternative doesn’t seem to be sufficient to qualify as a possible interpretation of a hyperbolic statement. This, as has already been noted in relation to the hybrid results obtained in Table 1, doesn’t pose a threat to the proposed conjecture: (5) is compatible with the observation that not all scalar alternatives are hyperbole-friendly. It is left for further research to determine what other constraint(s)—in addition to the one articulated in (5)—is/are needed to provide a conclusive solution to hyperbole’s weakening problem.

It’s also worth noting that a theory of hyperbole should do more than just solving hyperbole’s weakening problem: indeed, a theory of hyperbole should explain (i) why a hyperbolic statement is weakened the way it is as well as (ii) where the expressive/evaluative aspect of hyperbole comes from. This is what Kao et al. (2014) tried to do—and they should be praised for trying to do that (ultimately, that is what is needed). Here we have attempted something considerably less ambitious than them: a generalisation that may contribute to answering (i).

Finally, we would like to draw attention to a set of possible counterexamples to the proposed conjecture. Take, for example, the sentence in (15).

(15) John is a tower.

It seems uncontroversial that (15) can be interpreted as meaning that John is quite a lot taller than average. It's not clear to me, however, that (15) does have a scalar alternative that could be recruited to generate such an interpretation. Let's assume that as a matter of fact it doesn't: then, should we view (15), or similar examples, e.g., 'John is a giraffe', 'John is a pyramid', as counterexamples to the proposed conjecture?

We are not persuaded that we should. Indeed, there are at least two coherent ways of thinking about (15) that are (or can be made) compatible with (5). One possibility is that 'John is a tower' is a metaphor, and hence a manifestation of a different empirical phenomenon. Another possibility is that 'John is a tower' is both a metaphor and a hyperbole²⁸: 'John is a tower', a category mistake if John is taken to be a person, is first interpreted metaphorically—to simplify things, assume that the metaphor-computing mechanism involves replacing 'a tower' by '(at least) 30m tall'; then, 'John is (at least) 30m tall', which is almost certain to be false, can be interpreted hyperbolically as meaning, for example, 'John is (at least) 2m tall', which, under standard assumptions, is a scalar alternative of 'John is (at least) 30m tall'. (See §3.1 for some thoughts as to why 'John is (very) tall' would typically be preferred over 'John is (at least) 2m tall' as a paraphrase of (15)'s hyperbolic interpretation).

Of course, these two ways of looking at (15) make sense provided that the classical picture, according to which hyperbole and metaphor are distinct phenomena (which may or may not co-occur), is on the right track. If this picture turns out not to be correct, then (15) could be regarded as a problem for the proposed conjecture.

4 Final thoughts

Is hyperbole a scalar inference? The answer to this question depends on one's working definition of 'scalar inference'. What seems likely is that scalar implicatures and hyperbolic statements share a common element—namely, reliance on scalar alternatives.

²⁸ The view that hyperbole and metaphor can co-occur is already found in Aristotle. 'Well-liked hyperboles are also metaphors; for example, of a man with a black eye, — "You would have thought him a basket of mulberries"; for his face is somewhat purple, but there is much exaggeration.' (Aristotle [ca. 322 BCE] 2007: 225).

In this paper, we have tried to avoid as much as possible the issue of what it is that makes a sentence a scalar alternative of another sentence—suffice is to say that this is a hard problem for which a fully general solution doesn't yet exist (for discussion, see Breheny et al. 2018). This lack of a general solution, however, does not prevent us from agreeing on whether a given sentence should count (or shouldn't count) as a scalar alternative of another—for example, we agree that 'Most *blah*' should count as a scalar alternative of 'All *blah*' because 'Most *blah*' has the scalar implicature 'Not all *blah*'—and, if the conjecture advanced here is correct, because 'All *blah*' can be interpreted hyperbolically as meaning 'Most *blah*'.

Moving forwards, the research question that we find most exciting is the following: assuming that the proposed conjecture is correct, why is it that not all scalar alternatives can serve as landing site for the hyperbole inference—for example, why is it that 'p or q' has the implicature 'not(p or q)', but 'p and q' can't be interpreted hyperbolically as meaning 'p or q'? Answering this question in a principled way may contribute not only to better understanding how hyperbole works, but also to shedding light on the elusive nature of alternatives.

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